

TOPOLOGY

Topological Space:

A non-empty set X with a collection τ of subsets of X is called topological space (X, τ) if

- (i) $\phi, X \in \tau$
- (ii) Finite intersection of members of τ is also a member of τ .
- (iii) Union of an arbitrators family of members of τ is also a member of τ .

The following table shows the information of some topological spaces with their topology.

Set	Topology
R = set of Real number	(Metrizible topology) $\tau = \{A: A \text{ is an open set in metric space } (R, d), \text{ where } d \text{ is a usual metric on } R\}$
C = set of complex numbers	(Metrizible topology) $\tau = \{A: A \text{ is an open set in metric space } (C, d), \text{ where } d \text{ is a usual metric on } C\}$
N = set of natural numbers	$\tau = \{\phi, A_1, A_2, A_3, \dots\}$ Where $A_n = \{n, n+1, n+2, \dots\}$ for each $n \in N$.
X = non-empty set	$\tau = \{\phi, X\}$ (Indiscrete topology)
X = non-empty set	$\tau = P(X)$ $= \{\text{All subsets of } X\}$ (Discrete topology)
X = infinite set	$\tau = \phi \cup \{A: A \subseteq X \text{ such that } A' \text{ is finite}\}$ (Cofinite space) Note: If X is finite set then cofinite space is a simply a discrete space.
X = infinite set	$\tau = \phi \cup \{A: A \subseteq X \text{ such that } A' \text{ is countable}\}$ (Co-countable space)
$X = \{a\}$	$\tau = \{\phi, X\},$
$X = \{a, b\}$	$\tau = \{\phi, X\},$ $\tau = \{\phi, \{a\}, X\},$ $\tau = \{\phi, \{b\}, X\}$ $\tau = \{\phi, \{a\}, \{b\}, X\}$ Note: Total number of topologies on two elements set are 4.
$X = \{a, b, c\}$	$\tau = \{\phi, X\},$ $\tau = \{\phi, \{a\}, \{b\}, \{a, b\}, X\},$

	$\tau = \{\phi, \{b\}, \{b, c\}, X\}$ Note: Total number of topologies on three elements set are 29.
$X = \{a, b, c, d\}$	$\tau = \{\phi, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}, \{a, c, d\}, X\}$ Note: Maximum number of topologies on four elements set can be 2^{14} .

Open And Closed Sets:

Let (X, τ) be a topological space. A subset $A \subseteq X$ is open if $A \in \tau$. And A subset $B \subseteq X$ is closed if its complement $B' = X - B$ is open.

Weaker And Stronger Topologies:

Let X be nonempty set. Let τ_1 and τ_2 be two topologies on X such that $\tau_1 \subset \tau_2$. Then τ_1 is called weaker (smaller, coarser) topology and τ_2 is called stronger (larger, finer) topology on X .

Note: Let X be set with more than one point. Then discrete topology on X is the finer while indiscrete topology on X is the coarsest topology on X .

Base for Topology:

Let (X, τ) be topological space. A **subcollection** B of members of τ is called a basis (or base) for τ iff each open subset of X is a union of members of B .

The following table shows the information of some topological spaces with their bases.

Topology	Base
$(\mathbb{R}, \tau)$ Where τ = metrizable topology	$B = \{A: A \text{ is an open interval}\}$
$(\mathbb{R}^2, \tau)$ Where τ = metrizable topology	$B = \{A: A \text{ is an open rectangle in } \mathbb{R}^2\}$
(X, τ) Where τ = discrete topology	$B = \{A: A \text{ is singleton subset of } X\}$
$X = \{a, b, c, d\}$ $\tau = \{\phi, \{a\}, \{b\}, \{a, b\}, \{b, c\}, \{b, d\}, \{a, b, c\}, \{a, b, d\}, \{b, c, d\}, X\}$	$B_1 = \{\{a\}, \{b\}, \{b, c\}, \{b, d\}\}$ and $B_2 = \{\phi, \{a\}, \{b\}, \{b, c\}, \{b, d\}, X\}$ are two basis of τ .

Subspace with Relative Topology:

Let (X, τ) be a topological space. Let Y be any non-empty subset of X . Then (Y, τ_Y) is called subspace of X if τ_Y is a relative topology on Y .

i.e. $\forall A \in \tau_Y$ there exists $B \in \tau$ such that $A = Y \cap B$.

The following table shows the information of some topological spaces with their subspaces.

Topological space	Subspace
$(\mathbb{R}, \tau)$ Real line with Metrizible topology	$(\mathbb{N}, \tau_{\mathbb{N}})$ Natural number set with $\tau_{\mathbb{N}} = P(\mathbb{N})$ $(\tau_{\mathbb{N}}$ is a discrete topology).
(X, τ) where $X = \{a, b, c, d, e\}$ $\tau = \{\phi, \{a\}, \{b\}, \{a, b\}, \{b, c\}, \{b, d\}, \{b, e\}, \{a, b, c\}, \{a, b, d\}, \{b, c, d\}, \{b, c, e\}, \{b, d, e\}, \{a, b, c, d\}, \{a, b, c, e\}, \{a, b, d, e\}, \{b, c, d, e\}, X\}$	$Y = \{a, b, d\}$ $\tau_Y = \{\{a\}, \{b\}, \{a, b\}, \{b, d\}, \{b, d\}, Y\}$

Homeomorphism:

Let (X, τ_X) and (Y, τ_Y) be two topological spaces, A mapping $f: X \rightarrow Y$ is called homeomorphism (topological mapping) if

- (i) f is bijective
- (ii) f is continuous
- (iii) f^{-1} is continuous

And if $f: X \rightarrow Y$ is homeomorphism then X and Y are called homeomorphic (topologically equivalent).

The following table shows the information of some homeomorphic spaces with their homeomorphism.

Homeomorphic Space	Homeomorphis
$\mathbb{R} \cong (-1, 1)$ Where topologies on both spaces are metrizable topology of $\mathbb{R}$.	$f: \mathbb{R} \rightarrow (-1, 1)$ $f(x) = \frac{x}{1 + x }$ $\forall x \in \mathbb{R}$
$(-1, 1) \cong (a, b)$ Where topologies on both spaces are metrizable topology of $\mathbb{R}$.	$f: (-1, 1) \rightarrow (a, b)$ $f(x) = \frac{b+a}{2} + \frac{b-a}{2}$ $\forall x \in (-1, 1)$
$X \cong Y$ Where $X = \{a, b, c\}$, $\tau_X = \{\phi, \{c\}, \{a, b\}, X\}$ $Y = \{1, 2, 3\}$, $\tau_Y = \{\phi, \{3\}, \{1, 2\}, Y\}$	$f: X \rightarrow Y$ $f(a) = 1, f(b) = 2, f(c) = 3.$

To-Space:

To topological space (X, τ) is said to be To-space if for any two points $a, b, \in X$, there exists at least one open set which contains one of the points but not the other.

 T_1 -Space:

A topological space (X, τ) is said to be T_1 -space if for any two distinct points a, b there are open sets U and V which contains one but not the other.

 T_2 -Space (Hausdoff Space):

A topological space is said to be T_2 -space if for any two points $a, b \in X$, there exists open sets U and V such that $a \in U, b \in V$ and $U \cap V = \phi$.

Regular Space:

A topological space (X, τ) is said to be a regular space if for any closed set A and any point $x \in X$ such that $x \notin A$, there are open sets U and V such that

$$x \in U, A \subseteq V \text{ and } U \cap V = \phi$$

 T_3 -Space:

A regular T_1 -space is called T_3 space.

Completely Regular Space:

A topological space (X, τ) is said to be completely regular if, for any closed set A in X and a point $x \in X$ such that $x \notin A$, there exists a continuous function

$$f: X \rightarrow [0, 1]$$

such that

$$f(x) = 0 \text{ and } f(A) = 1$$

 $T_{3\frac{1}{2}}$ - Space (Tychonoff Space):

A completely regular T_1 -space is said to be a tychonoff space of $T_{3\frac{1}{2}}$ space.

Normal Space:

A space (X, τ) is said to be a normal space if for any two closed disjoint subsets $A, B \subseteq X$ there are open sets U and V such that $A \subseteq U, B \subseteq V$ and $U \cap V = \phi$

Following table shows the information of sepration spaces and some of their examples.

Space	Example
T_0 -Space	Every metric space with metrizable topology.
	$X = \{a, b\}$ with $\tau = \{\emptyset, \{a\}, X\}$ (Sierpinski space)
	Every discrete space
	Indiscrete space
	Every co-finite topological space.
T_1 -Space	Every metric space with metrizable topology
	Every discrete space with at least two points.
	Every co-finite topological space.
T_2 -Space	Every metric space with metrizable topology
	Every discrete space with at least two points.
Regular-Space	Every metric space with metrizable topology
	Every discrete space with at least two points
	$X = \{a, b, c\}$ with $\tau = \{\emptyset, \{a\}, \{b, c\}, X\}$
	Every indiscrete space
T_3 -Space	Every metric space with metrizable topology.
	Every discrete space
Completely Regular Space	Every metric space with metrizable topology
$T_{3\frac{1}{2}}$ Space (Tychonoff Space)	Every metric space with metrizable topology.
Normal Space	Every metric space with metrizable topology.
	Every discrete space with at least two points.
	$X = \{a, b, c\}$ with $\tau = \{\emptyset, \{a\}, \{b\}, \{a, b\}, X\}$
T_4 -Space	Every metric space with metrizable topology.
	Every discrete space with at least two points.

Compact Space:

A topological space (X, τ) is said to be compact if every open cover of X has a finite sub-cover. For example;

1. Any set X with topology τ such that no of elements in τ are finite is complete.
2. Any finite set X with topology τ is compact
3. An infinite set X with co-finite topology τ is compact.

4. $\mathbb{R}$ with usual topology is not compact.

Connected Space:

A topological space X is said to be connected if there does not exist a pair A, B of non-empty disjoint open subset of X such that $X = A \cup B$.

Otherwise space is called disconnected. For example;

1. Every indiscrete space is connected.
2. Any discrete space with more than one point is disconnected.
3. Any infinite space with co-finite topology is connected.
4. $\mathbb{R}$ with usual topology is connected
5. $\mathbb{Q}$ as a subspace of $\mathbb{R}$ is disconnected.

Multiple Choice Questions

1. Let X be a non-empty set. Let collection τ of subsets of X is a topology on X . Then which of the following statement is true.
(a) The union of any number of member of τ belongs to τ
(b) The intersection of a finite number of members of τ belongs to τ
(c) $\phi, X \in \tau$
(d) All of these
2. Let (X, τ) be a topological space such that $A \in \tau$. Then A is
(a) subset (b) closed set (c) open set (d) All of these
3. Let τ_1 and τ_2 two topologies on a set X such that $\tau_1 \subseteq \tau_2$. Then which of the following statement is true.
(a) $\tau_1 \cap \tau_2$ is a topology on X (b) τ_1 is weaker (coarser) than τ_2
(c) τ_2 is stronger (finer) than τ_1 (d) All of these
4. Let X be a empty set. Then which of the following topology is the weakest topology on X .
(a) $\tau = \{\phi, X\}$ (indiscrete topology) (b) $\tau = P(X)$ (discrete topology)
(c) τ is a metric topology (d) None
5. Let (X, τ) be a topological space with discrete topology. Then τ is
(a) weakest topology on X (b) strongest topology on X
(c) middle topology on X (d) None
6. Let (X, d) be a metric space. Let $\tau = \{U \subseteq X \mid \text{for each } a \in U \text{ there is open ball } B(a; r) \subseteq U\}$. Then τ is a
(a) discrete topology (b) indiscrete topology
(c) metrizable topology (d) None
7. Let $X = \{x\}$. Then how many topologies can be made on X .
(a) 1 (b) 2 (c) 3 (d) 4
8. Let $X = \{x, y\}$. Then number of different topologies on X is
(a) 1 (b) 2 (c) 4 (d) 6
9. Let $X = \{x, y, z\}$. Then number of topologies on X is
(a) 1 (b) 14 (c) 8 (d) 29
10. Let $X = \{x, y, z\}$. Then which of the following is a topology τ on X .
(a) $\tau = \{\phi, X, \{z\}\}$ (b) $\tau = \{\phi, X, \{x\}, \{y\}, \{x, y\}\}$
(c) $\tau = \{\phi, X, \{x\}, \{z\}, \{x, y\}, \{x, z\}\}$ (d) All of these
11. Let $X = \{w, x, y, z\}$. The maximum number of topologies on X can be
(a) 166 (b) 2^4 (c) 16323 (d) 2^{14}
12. Let $X = \{a, b, c\}$ and $\tau = \{\phi, X, \{a\}, \{b\}, \{a, b\}\}$ be a topology on X . Then which of the following set is not closed in X .
(a) $\{b, c\}$ (b) $\{c\}$
(c) $\{a, c\}$ (d) All of these

13. Let (X, τ) be a topological space. Then which of the following set is closed as well as open (clopen) set.
 (a) $\{a\}$ (b) $\{a, b\}$ (c) X (d) $\{c\}$
14. Let $X = \{a, b, c\}$. Then in which of the following topology on X has all subsets are clopen.
 (a) $\tau = \{\emptyset, X\}$ (b) $\tau = P(X)$
 (c) $\tau = \{\emptyset, X, \{a\}, \{b, c\}\}$ (d) All of these
15. Let (X, τ) be a topological space and (Y, τ') be a subspace of (X, τ) . Then for $A \subseteq \tau$ and for $B \subseteq \tau'$ which of the following can be possible
 (a) $A \cap X = B$ (b) $B \cap Y = A$
 (c) $A \cap Y = B$ (d) None
16. Which of the following statement is correct
 (a) subspace of a discrete space is discrete
 (b) subspace of a indiscrete space is indiscrete
 (c) If A is a subspace of B and B is a subspace of C , then A is a subspace of C .
 (d) All of these
17. Let $(X, \leq)$ be partially ordered set. For any $x \in X$ let $S(x) = \{y \in X \mid x \leq y\}$ and $\tau = \{U \subseteq X \mid \text{whenever } x \in U \text{ then } S(x) \subseteq U\}$. Then τ is a
 (a) metrizable topology (b) right order topology
 (c) indiscrete topology (d) None
18. Let (X, τ) be a topological space and B is a sub-collection of τ such that $\forall A \in \tau, A = \text{union of members of } B$. Then B is
 (a) topology on X (b) relative topology on X
 (c) base for topology τ on X (d) None
19. Let (X, τ) be a discrete topological space. Then which of the following is base for topology τ .
 (a) $B = \{\{x\} \mid x \in X\}$ (b) $B = \{X\}$
 (c) $B = \{\emptyset, X\}$ (d) None
20. Which of the following collection is a base for indiscrete topology on X .
 (a) $B = \{\{x\} \mid x \in X\}$ (b) $B = \{X\}$
 (c) $B = \{\emptyset, X\}$ (d) None
21. Which of the following collection is a base for metric topology on X
 (a) $B = \{\{x\} \mid x \in X\}$ (b) $B = \text{set of all open balls in } X$.
 (c) $B = \{\emptyset, X\}$ (d) None
22. If $S(x) = \{y \in X \mid x \leq y\}$, then which of the following is a base for right order topology on X .
 (a) $B = \{\{x\} \mid x \in X\}$ (b) $B = \{S(x) \mid x \in X\}$
 (c) $B = \{\emptyset, X\}$ (d) None
23. Let $X = \{a, b, c, d\}$ and $\tau = \{\emptyset, \{a\}, \{b\}, \{a, b\}, \{b, c\}, \{b, d\}, \{a, b, d\}, \{a, b, c\}, \{b, c, d\}, X\}$ be a topology on X . Then which of the following is a base t .
 (a) $\{\{a\}, \{b\}, \{a, b\}\}$ (b) $\{\{a\}, \{b\}, \{a, b, c\}\}$
 (c) $\{\{a\}, \{b\}, \{b, c\}, \{b, d\}\}$ (d) $\{\{a\}, \{b, c\}, X\}$
24. Let (X, τ) be a topological space and let B be sub-collection of τ . If for any $x \in U \in \tau$ there is $A \in B$ such that $x \in A \subseteq U$. Then
 (a) B is a collection of open set (b) B is a neighbourhood base at x

- (c) B is a topology at x (d) None
25. In a metric space (X, d) which of the following is a neighbourhood base for any $x \in X$.
 (a) $\{B(x; r) \mid r \in \mathbb{R}, r > 0\}$ (b) $\{\{x\} \mid x \in X\}$
 (c) $\{\emptyset, \{x\}, X\}$ (d) None
26. Which of the following topological space X has a finite neighbourhood base at each of its point. That is for each $x \in X$, $B = \{\{x\}\}$ is neighbourhood base at x .
 (a) Discrete topological space (b) indiscrete topological space
 (c) real topological space (d) None
27. Let X be a non-empty set and $\tau = \{A \subseteq X \mid A' \text{ is finite}\} \cup \emptyset$ be a topology on X . Then τ is a
 (a) discrete topology (b) Indiscrete topology
 (c) co-finite topology (d) None
28. Let (X, τ) be a topological space and for each $x \in X$, there is a countable neighbourhood base at x . Then (X, τ) is called
 (a) 1st countable space (b) 2nd countable space
 (c) 3rd countable space (d) None
29. If there is a countable base for topology τ in a topological space (X, τ) . Then (X, τ) is called
 (a) 1st countable space (b) 2nd countable space
 (c) 3rd countable space (d) None
30. Which of the following statement is true
 (a) every 2nd countable space is also 1st countable space
 (b) the co-finite space is not 1st countable space
 (c) if X is uncountable, then X with discrete topology is not 2nd countable space
 (d) All of these
31. Let (X, τ) be a topological space. Let collection $S = \{U \mid U \in \tau\}$ such that $X = \bigcup_{\alpha} U_{\alpha}$, $\alpha \in I$. Then S is called
 (a) topology on X (b) open cover for X
 (c) partition for X (d) None
32. In a topological space (X, τ) , if every open cover has a countable subcover, then (X, τ) is called
 (a) separable space (b) countable space
 (c) Lindelof space (d) Category
33. Which of the following statement is correct
 (a) Every 2nd countable space is Lindelof.
 (b) Every 2nd countable space is separable
 (c) Every separable metric space is 2nd countable
 (d) All of these
34. Let (X, τ) , (Y, τ') be two topological spaces. A function $f: X \longrightarrow Y$ is continuous if
 (a) for each $x \in X$ if V is open set, and $f(x) \in V \subseteq Y$ then there is open set U , $x \in U \subseteq X$ such that $f(U) \subseteq V$.
 (b) for each open subset V in Y , $f^{-1}(V)$ is open in X .
 (c) for each closed subset V in Y , $f^{-1}(V)$ is closed in X
 (d) All of these

35. Let (X, τ_1) and (X, τ_2) be two topological space. A function $f: X \longrightarrow Y$ is continuous if
- (a) τ_1 is stronger than τ_2 (b) τ_1 is a discrete topology
(c) τ_2 is a indiscrete topology (d) All of these
36. Let (X, τ_1) & (Y, τ_2) be two topological space where $X = \{a, b, c\}$, $Y = \{1, 2, 3\}$
 $\tau_1 = \{\phi, X, \{b\}, \{a, b\}, \{b, c\}\}$
 $\tau_2 = \{\phi, Y, \{1\}\}$
 A function $f: X \longrightarrow Y$ is continuous if
- (a) $f(a) = 1, f(b) = 2, f(c) = 3$ (b) $f(a) = 2, f(b) = 3, f(c) = 1$
 (c) $f(a) = 2, f(b) = 1, f(c) = 3$ (d) $f(a) = 3, f(b) = 2, f(c) = 1$
37. Let $f: X \longrightarrow Y$ be a Homeomorphism. Then which of the following is correct.
- (a) f is bijective (b) f is continuous
(c) f^{-1} is continuous (d) All of these
38. Let X and Y be two topological space and $f: X \longrightarrow Y$ is a bijective function. Then which of the following is equivalent to other.
- (a) f is homeomorphism (b) $f(U)$ is open in Y iff U is open in X
 (c) for any $A \subseteq X$, $f\left(\overline{A}\right) \subseteq \overline{f(A)}$ (d) All of these
39. Which of the following subspace of $\mathbb{R}$ has discrete topology.
- (a) $\mathbb{Q}$ (b) $\mathbb{Q}'$ (c) $\mathbb{Z}$ (d) $[0, 1]$
40. Let (X, τ) be a topological space. If for any two points $a, b, \in X$, there exists $U \in \tau$ such that either $a \in U$ or $b \in U$ but not both. Then (X, τ) is
- (a) T_0 - Space (b) T_2 - Space
(c) T_1 - Space (d) T_3 - Space
41. Which of the following is a T_0 - Space.
- (a) Discrete space
(b) $X = \{a, b\}$ with Sierpinski topology $\tau = \{\phi, \{a\}, X\}$
(c) Real line $\mathbb{R}$ with usual topology
(d) All of these
42. Let (X, τ) be a topological space. If for any two distinct point $a, b \in X$ there are open sets U and V which contain one but not the other, then (X, τ) is
- (a) T_0 - Space (b) T_2 - Space
(c) T_1 - Space (d) T_3 - Space
43. Which of the following space is not T_1 - Space.
- (a) Discrete space
(b) Real line $\mathbb{R}$ with usual topology
(c) An infinite set X with Co-finite topology
(d) $X = \{a, b\}$ with Sierpinski topology $\tau = \{\phi, \{a\}, X\}$

44. Let (X, τ) be a T_1 – space. Then which of the following is correct.
- (a) Every singleton subset of X is closed. (b) Every finite subset of X is closed
(c) No finite subset of X has limit point (d) All of these
45. Let (X, τ) be a topological space. If for every two distinct points $a, b, \in X$ there is $U, V \in \tau$ such that $a \in U, b \in V$ and $U \cap V = \phi$. Then (X, τ) is
- (a) T_0 – Space (b) T_2 – Space (Hausdorff)
(c) T_1 – Space (d) T_3 – Space
46. Which of the following is not T_2 – space (Hausdorff)
- (a) Discrete space will at least two elements
(b) Every metric space
(c) Real line $\mathbb{R}$ with usual topology
(d) An infinite set X with cofinite topology
47. In a topological space X if for any closed set A and any point $x \notin A$, there are disjoint open sets U and V such that $x \in U, A \subseteq V$, then X is:
- (a) T_1 space (b) T_2 space
(c) Hausdorff space (d) Regular space
48. Which of the following is a regular space:
- (a) Discrete space (b) Indiscrete space
(c) Metrizable space (d) T_1 -space
49. In a topological space X , if for any closed set A and any point $x \notin A$, there is a continuous function $f : X \rightarrow [0, 1]$ such that $f(x) = 0$ and $f(A) = 1$, then X is called:
- (a) regular space (b) normal space
(c) completely regular space (d) none
50. Which of the following is completely regular space:
- (a) discrete space (b) indiscrete space
(c) metric space (d) Hausdorff space
51. Which of the following statement is correct:
- (a) A T_3 -space is a regular T_1 -space (b) A completely regular T_1 -space is Hausdorff
(c) A completely regular space is regular (d) All of these
52. A completely regular T_1 space is also called:
- (a) Tychonoff space (b) $T_{3\frac{1}{2}}$ -space
(c) (a) & (b) (d) none
53. In a topological space, if for any two disjoint closed set A and B there are disjoint open sets U and V such that $A \subseteq U$ and $B \subseteq V$, then X is called:
- (a) regular space (b) Tychonoff space
(c) normal space (d) regular space
54. Which of the following is normal space.
- (a) metric space (b) discrete space with atleast two points
(c) closed subspace of a normal space (d) All of these

55. Which of the following is T_4 -space.
- (a) regular T_2 -space (b) normal T_1 -space
(c) completely regular T_3 -space (d) T_2 -Hausdorff space
56. Let (X, τ) be a topological space such that every open cover of X has a finite subcover, then X is called:
- (a) complete (b) connected
(c) compact (d) coverless
57. Which of the following is compact:
- (a) cofinite space (b) indiscrete space
(c) discrete space X , where X is finite (d) all of these
58. Which of the following statement is true.
- (a) compactness is a topological property
(b) compact subset of a Hausdorff space is closed
(c) closed subset of a compact space is compact
(d) all of these
59. Which of the following is correct.
- (a) every compact Hausdorff space is normal
(b) every compact regular space is normal
(c) every compact T_1 -space is normal
(d) none
60. A set X is closed and bounded if X is:
- (a) compact subset of $\mathbb{R}^n$ (b) $[0, 1]$
(c) compact subset of $\mathbb{R}$ (d) all of these
61. If every infinite subset of X has limit point in X then X satisfies:
- (a) Lagrange's identity (b) Bolzano-Weierstrass property
(c) Schwarz inequality (d) none
62. Which of the following is correct:
- (a) every sequentially compact space is countably compact
(b) every countably compact space satisfies Bolzano-Weierstrass property
(c) every sequentially compact space satisfies Bolzano-Weierstrass property
(d) all of these
63. Let X be a countably compact metric space, then X is:
- (a) second countable (b) totally bounded
(c) sequentially compact (d) all of these
64. Let A and B disjoint non-empty open subsets of X such that $X = A \cup B$, then X is:
- (a) compact (b) connected
(c) disconnected (d) complete
65. Which of the following space is connected:
- (a) indiscrete space (b) cofinite space
(c) real line $\mathbb{R}$ (d) all of these

66. Which of the following is totally disconnected space.
- | | |
|-----------------------|------------------|
| (a) discrete space | (b) $\mathbb{Q}$ |
| (c) $\mathbb{Q}'$ (d) | all of these |

Key To MCQ Of Topology

1	d	2	c	3	d	4	a	5	b	6	c
7	a	8	c	9	d	10	d	11	d	12	d
13	d	14	d	15	c	16	d	17	b	18	c
19	a	20	b	21	b	22	b	23	c	24	b
25	a	26	a	27	c	28	a	29	b	30	d
31	b	32	c	33	d	34	d	35	d	36	c
37	d	38	d	39	c	40	a	41	d	42	b
43	d	44	d	45	c	46	d	47	d	48	b
49	c	50	c	51	d	52	c	53	c	54	d
55	b	56	c	57	d	58	d	59	a	60	d
61	b	62	d	63	d	64	c	65	a	66	d